BEVFormer: Learning Bird's-Eye-View Representation From LiDAR-Camera via Spatiotemporal Transformers

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***Abstract*—Multi-modality fusion strategy is currently the de-facto most competitive solution for 3D perception tasks. In this work, we present a new framework termed BEVFormer, which learns unified BEV representations from multi-modality data with spatiotemporal transformers to support multiple autonomous driving perception tasks. In a nutshell, BEVFormer exploits both spatial and temporal information by interacting with spatial and temporal space through predefined grid-shaped BEV queries. To aggregate spatial information, we design spatial cross-attention that each BEV query extracts the spatial features from both point cloud and camera input, thus completing multi-modality information fusion under BEV space. For temporal information, we propose temporal self-attention to fuse the history BEV information recurrently. By comparing with other fusion paradigms, we demonstrate that the fusion method proposed in this work is both succinct and effective. Our approach achieves the new state-of-the-art 74.1% in terms of NDS metric on the nuScenes test set. In addition, we extend BEVFormer to encompass a wide range of autonomous driving tasks, including object tracking, vectorized mapping, occupancy prediction, and end-to-end autonomous driving, achieving outstanding results across these tasks.**

***Index Terms*—Autonomous driving, bird's-eye-view, 3D object detection, map segmentation, transformer, LiDAR-camera.**

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The code is released at https://github.com/fundamentalvision/BEVFormer.

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I. Introduction

PERCEPTION in 3D space is critical for various applications such as autonomous driving, robotics, *etc*. LiDAR and cameras are the most widely used sensors of 3D perception systems [1], [2]. Different sensors have discrepant perceptual information about the surrounding environment. Point clouds from LiDAR explicitly provide accurate spatial information in the scene, while RBG images captured from cameras have a wealth of semantic information [3], [4], [5], [6]. Due to the complementary characteristics of different modalities, sensor fusion is crucial to building an accurate and safe autonomous driving perception system [7], [8], [9], [10]. In this work, *we aim to extend BEVFormer, a pure camera-based 3D detector of our conference version, into a unified multi-modality perception framework.* Owing to view discrepancies and semantic inconsistencies between different modalities, it is necessary to transform data from diverse modalities into a shared space for effective feature fusion. In this paper, we categorize the mainstream multi-modality fusion methods into four types: (a) Vision-Centric Fusion, (b) LiDAR-Centric Fusion, (c) Independent Fusion, (d) Interactive Fusion.

LiDAR-centric fusion is to extend the LiDAR-based methods by incorporating 2D image information to enhance point clouds or augment the point cloud properties [6], [11]. Although this approach is simple and effective, it primarily focuses on enhancing LiDAR-based methods and does not fully exploit the rich semantic information available in multi-view images. Unlike methods that use image information to enhance the semantic information of point clouds, vision-centric approaches such as VCD [12] employ point clouds to produce accurate depth information to guide the generation of image-based BEV features. However, the drawback of this method is that it can only utilize the depth information provided by point clouds and is unable to effectively leverage other information contained within the point clouds.

Since the bird's-eye-view (BEV) representations have gained popularity, independent fusion methods [4], [5], [10] propose transforming both 2D images and point cloud data are transformed into a shared BEV space, where simple fusion techniques such as concatenation and element-wise addition are applied. However, there are two primary challenges associated with this approach. (1) The accuracy of depth estimation and camera calibration greatly influences the quality of the BEV

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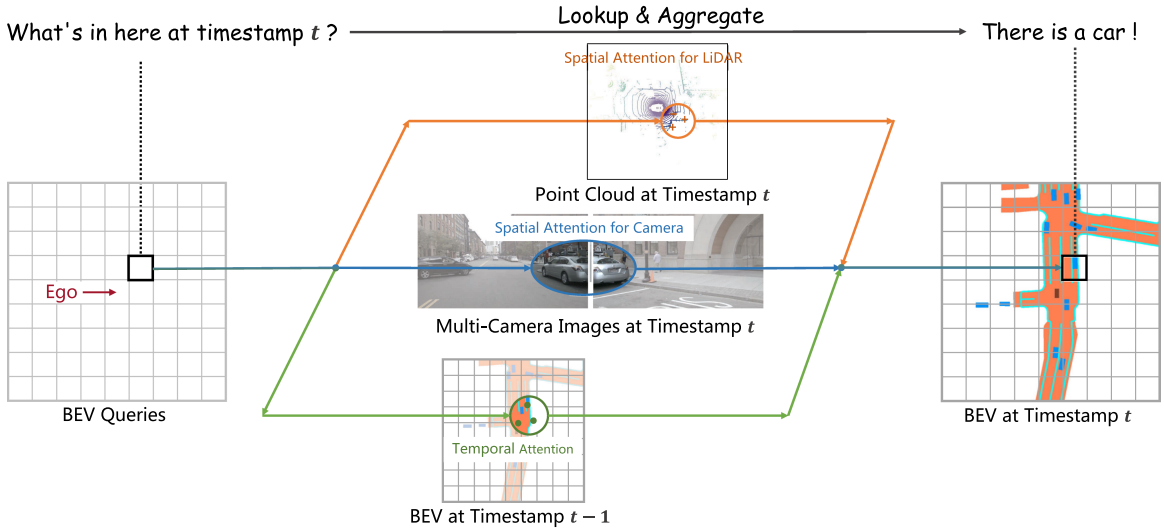


Fig. 1. We propose *BEVFormer*, a paradigm for autonomous driving that applies both transformer and temporal structures to generate bird's-eye-view (BEV) features from LiDAR-camera inputs. BEVFormer leverages queries to lookup spatial/temporal space and aggregate spatiotemporal information correspondingly, hence benefiting stronger representations for perception tasks.

representation derived from 2D images, potentially introducing errors into the resulting representation. (2) Despite having the same spatial view, the cross-modality BEV features may exhibit a domain gap in their contextual features. The presence of these inconsistencies in the context features creates difficulties in effectively fusing multi-modality data. Considering the shortcomings of existing fusion methods, in this paper, we propose an innovative interactive fusion approach that can extend the conference version of BEVFormer into a multi-modality fusion method with a simple yet effective strategy. Specifically, for multi-modality spatial aggregation, different from the simplistic fusion methods employed in previous works [4], [5], the upgraded BEVFormer leverages a unified multi-modality spatial attention module. The BEV query obtains multi-modality information by interacting with different modalities and completes the interaction and fusion of multiple modalities at the query level. Since BEVFormer gradually fuses multi-modality spatial information through multiple interactions aggressively, the BEV queries that integrate multi-modality information can obtain more accurate modal information in the next stage of spatial interaction with specific modalities.

In addition to multi-modality spatial aggregation, we explore an orthogonal approach to enhance the BEV representation at the temporal dimension. Temporal information is crucial in human visual perception for inferring object motion states and identifying occluded objects [13], [14], [15], [16], [17]. However, existing multi-modality 3D detection methods [4], [5], [7], [10] rarely exploit temporal information. While stacking multiple sweeps of the point cloud can partially encode temporal context [1], [10], it fails to capture long-term dependencies and neglects rich semantic information from the camera stream. To mine temporal information in all modalities and build longer temporal dependencies, we fuse all modalities into a shared BEV representation, creating a bridge for propagating fusion information across the temporal dimension. Nonetheless, challenges arise due to the time-critical nature of autonomous driving and the rapid changes occurring in the scene. Stacking BEV features from different timestamps introduces computational costs and interference. Inspired by recurrent neural networks (RNNs) [18], [19], we leverage BEV features to recurrently deliver temporal information, akin to RNN hidden states.

Combining the aforementioned techniques, we present a transformer-based BEV encoder, termed *BEVFormer*, which can effectively aggregate spatiotemporal features from multi-modality data from LiDAR and camera sensors and history BEV features. The BEV features generated from the BEVFormer can simultaneously support multiple 3D perception tasks such as 3D object detection and map segmentation, which are crucial for autonomous driving systems. As shown in Fig. 1, our BEVFormer contains three key designs, which are (1) *grid-shaped BEV queries* to accurately fuse spatial and temporal features via attention mechanisms flexibly, (2) *multi-modality spatial cross-attention* to aggregate the spatial features from different modalities, and (3) *temporal self-attention* to extract temporal information from history BEV features with negligible computational overhead, improving velocity estimation for moving objects and enhancing the detection of heavily occluded objects. With the unified BEV features, our model can collaborate with different task-specific heads, such as Deformable DETR [20] and mask decoder [21], for end-to-end 3D object detection and map segmentation.

This work presents a significant extension compared to our previous conference version [22], specifically in the development of a unified multi-modality fusion framework. In the conference version, the proposed method was limited to handling only the camera modality, resulting in performance and compatibility limitations. In this work, we extend BEVFormer to a unified perception framework capable of supporting both LiDAR and camera. This enhancement enables significantly more accurate autonomous driving perception, as well as the

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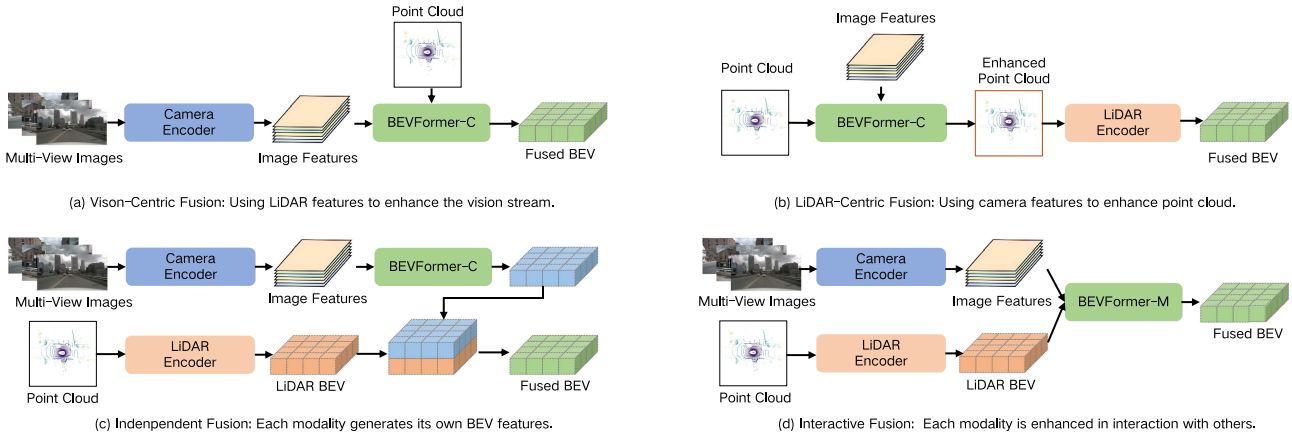


Fig. 2. To extend BEVFormer from a vision-only model in the conference version to a multi-modality model, we designed and compared the performance of various fusion strategies. (a) Version-A: Utilizing LiDAR information to enhance vision-centric BEV generation. In this implementation, We use the point cloud to keep valid 3D sampling points of BEVFormer. (b) Enhancing point cloud features using image information, exemplified by PointPainting [6] and PointAugmenting [11]. For each point cloud, we extract the corresponding image features to serve semantic clues in this implementation. (c) Independently generating BEV features for each modality and then fusing them through simple feature stacking, as demonstrated by BEVFusion-MIT [4] and BEVFusion-PKU [5]. This implementation is basically following BEVFusion-MIT. (d) Our proposed interactive modality fusion approach, where each modality is interlinked and mutually enhances the others, thereby achieving superior performance.

ability to process arbitrary single-modality inputs. To more accurately compare the strengths and weaknesses of different fusion methods, we also implemented the aforementioned three other fusion approaches based on the BEVFormer of the conference version, as shown in Fig. 2. By comparing different fusion methods within the same framework, we demonstrated the superiority of our proposed interactive fusion strategy. In addition, we also extended BEVFormer to more downstream tasks in autonomous driving, demonstrating the effectiveness of our approach in object tracking, occupancy prediction, vectorized map construction, and end-to-end autonomous driving.

Our main contributions are as follows:

- We propose BEVFormer, a multi-modality spatiotemporal transformer encoder that projects LiDAR-camera stream input to BEV representations. Leveraging the unified BEV features, our model excels in performing multiple high-accuracy perception tasks for autonomous driving, including 3D detection, 3D object tracking, map segmentation, vectorized map construction, occupancy prediction, and end-to-end autonomous driving.
- We design learnable BEV queries along with multi-modality spatial cross-attention and temporal self-attention layers to lookup spatial features from surrounding cameras, LiDAR, and temporal features from history BEV, respectively, and then aggregate them into unified BEV features. To more rigorously prove the superiority of our fusion method, we re-implemented mainstream multi-modality fusion methods based on our conference version of BEVFormer for comparison.
- We evaluate our BEVFormer on the challenging nuScenes [3] benchmarks and consistently achieve improved performance compared to the prior arts. For the object detection task, with camera-LiDAR input, BEVFormer achieves the state-of-the-art 74.1% NDS (+3.9% over the LiDAR baseline), which dominates the effectiveness of sensor fusion. For end-to-end autonomous driving, BEVFormer achieves a collision rate of 0.68% and outperforms UniAD [23] with 0.09%. We hope this straightforward and strong framework can serve as a new baseline for following 3D perception tasks.

II. Related Work

A. Transformer-Based 2D Perception

Recently, a new trend is to use transformers to reformulate detection and segmentation tasks [20], [21], [24]. DETR [24] uses a set of object queries to generate detection results by the cross-attention decoder directly. However, the main drawback of DETR is the long training time. Deformable DETR [20] solves this problem by proposing deformable attention. Different from vanilla global attention in DETR, the deformable attention interacts with local regions of interest, which only samples K points near each reference point and calculates attention results, resulting in high efficiency and significantly shortening the training time. The deformable attention mechanism is calculated by

$$\mathrm{DAttn}(q,p,x)=\sum_{i=1}^{N_{\text{head}}}\mathcal{W}_i\sum_{j=1}^{N_{\text{key}}}\mathcal{A}_{ij}\cdot\mathcal{W}_i'x\left(p+\Delta p_{ij}\right), \quad (1)$$

where q, p, x represent the query, reference point and input features, respectively. i indexes the attention head, and N_{head} denotes the total number of attention heads. the sampled keys is j, and the number of total sampled keys of each head is N_{key}. $W_i \in \mathbb{R}^{C\times(C/H_{\text{head}})}$ and $W_i' \in \mathbb{R}^{(C/H_{\text{head}})\times C}$ are the learnable weights, where C is the feature dimension. $A_{ij} \in [0,1]$ is the predicted attention weight, and is normalized by $\sum_{j=1}^{N_{\text{key}}} A_{ij} = 1$. $\Delta p_{ij} \in \mathbb{R}^2$ are the predicted offsets to the reference point p. $x(p+\Delta p_{ij})$ represents the feature at location $p+\Delta p_{ij}$, which is extracted by bilinear interpolation as in Dai et al. [25]. In this

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work, we extend the deformable attention to 3D perception tasks, to efficiently aggregate both spatial and temporal information.

B. Camera-Based 3D Perception

Previous 3D perception methods typically perform 3D object detection or map segmentation tasks independently. For the 3D object detection task, early methods are similar to 2D detection methods [26], [27], [28], [29], which usually predict the 3D bounding boxes based on 2D bounding boxes. Wang et al. [30] follows an advanced 2D detector FCOS [31] and directly predicts 3D bounding boxes for each object. DETR3D [32] projects learnable 3D queries in 2D images, and then samples the corresponding features for end-to-end 3D bounding box prediction without NMS post-processing. PETR [33], [34] introduces the vanilla attention mechanism and 3D positional encoding to enhance the generality of the model architecture. Another solution is to transform image features into BEV features and predict 3D bounding boxes from the top-down view. Methods transform image features into BEV features with the depth information from depth estimation [35] or categorical depth distribution [36], [37], [38]. OFT [39] and ImVoxelNet [40] project the predefined voxels onto image features to generate the voxel representation of the scene. M ^{2} BEV [41] further explored the feasibility of simultaneously performing multiple perception tasks based on BEV features. BEVDepth [42] model introduces point clouds during the training process to generate accurate depth maps for supervising the depth prediction module, thereby enhancing the model's geometric perception capabilities. Based on BEVDepth, BEVStereo [43] utilizes temporal multi-view stereo to benefit the depth estimation. Recent methods [44], [45], [46] explore integrating longer temporal information to enhance its geometric perception capabilities. Although camera-based methods have made significant progress recently, there is still a certain gap compared to LiDAR-based methods due to the lack of accurate geometric information in the visual modality.

C. LiDAR-Based 3D Perception

Detection in BEV space is one of the most popular schemes among LiDAR-based detectors. For BEV representation generation, raw point clouds can be quantized into voxels or pillars to form a regular grid shape [1], [47]. In this work, we use the LiDAR detector TransFusion-L [10] as our LiDAR stream and use voxel representation for BEV generation. There are also point-based methods (PointRCNN [48], STD [49], 3DSSD [50]) which perceive 3D objects based on the extracted point cloud features.

D. Multi-Modality 3D Perception

Due to the mutual complementarity between camera and LiDAR data, sensor-fusion approaches have dominated the leaderboards for autonomous driving perception. By capitalizing on the abundant semantic information present in camera data and the precise geometric information provided by LiDAR data, the adoption of a multi-modality fusion solution emerges as the most promising approach among existing methods for autonomous driving perception. Existing approaches can be classified into three categories: point-level fusion, proposal-level fusion, and BEV-level fusion. Point-level fusion methods usually map image semantic features onto LiDAR points to augment the point properties or density the point clouds (PointPainting [6], PointAugmenting [11], MVP [51], FusionPainting [52], AutoAlign [9], DeepFusion [53]). Proposal-level fusion methods usually generate instance-level proposals from LiDAR data to query the associated 2D images for enhancing object features. This paradigm only focuses on foreground objects, while ignoring the rich semantic information in the background. (TransFusion [10], FUTR3D [8]).

As another line of work, BEV-level fusion methods generate BEV representation of each modality and perform cross-modality fusion with these BEV features. The simplest strategy is just concatenating all BEV features along the channel dimension [5]. The weakness of these methods lies in the separate generation of BEV representations for different modalities, resulting in a lack of effective interaction between modalities. Deepinteraction [7] introduces two types of fusion representations: LiDAR-to-image and image-to-LiDAR, aiming to facilitate the fusion between different modalities. However, the requirement of using these two different representations for task heads limits the flexibility of the model. In this work, we propose a multi-modality spatial cross-attention that fuses features from different modalities with the unified BEV queries. Through iterative optimization, the BEV query can acquire more accurate multi-modality information. In this manner, features from different modalities will be fused to facilitate the querying of specific modalities in the subsequent fusion stage. In addition, we also propose a multi-modality temporal fusion mechanism, which differs from performing temporal fusion within specific individual modalities. Temporal fusion within the fused modality simplifies the design of temporal fusion while allowing the temporal fusion stage to leverage richer feature information.

III. BEVFormer

Converting multi-camera images and point clouds to bird's-eye-view (BEV) features can provide a unified surrounding environment representation for various autonomous driving perception tasks. In this work, we present a new transformer-based framework for BEV generation, which can effectively aggregate spatiotemporal features from the surround-view camera, LiDAR data and history BEV via attention mechanisms.

A. Overall Architecture

As illustrated in Fig. 3, BEVFormer has 3 encoder layers, each of which follows the conventional structure of transformers [54], except for three tailored designs, namely BEV queries, multi-modality spatial cross-attention, and temporal self-attention. Specifically, BEV queries are grid-shaped learnable parameters, which are designed to query features in BEV space from multi-camera views or point clouds via attention mechanisms. Multi-modality spatial cross-attention and temporal self-attention are attention layers working with BEV queries, which are used to lookup and aggregate spatial features from multi-view image

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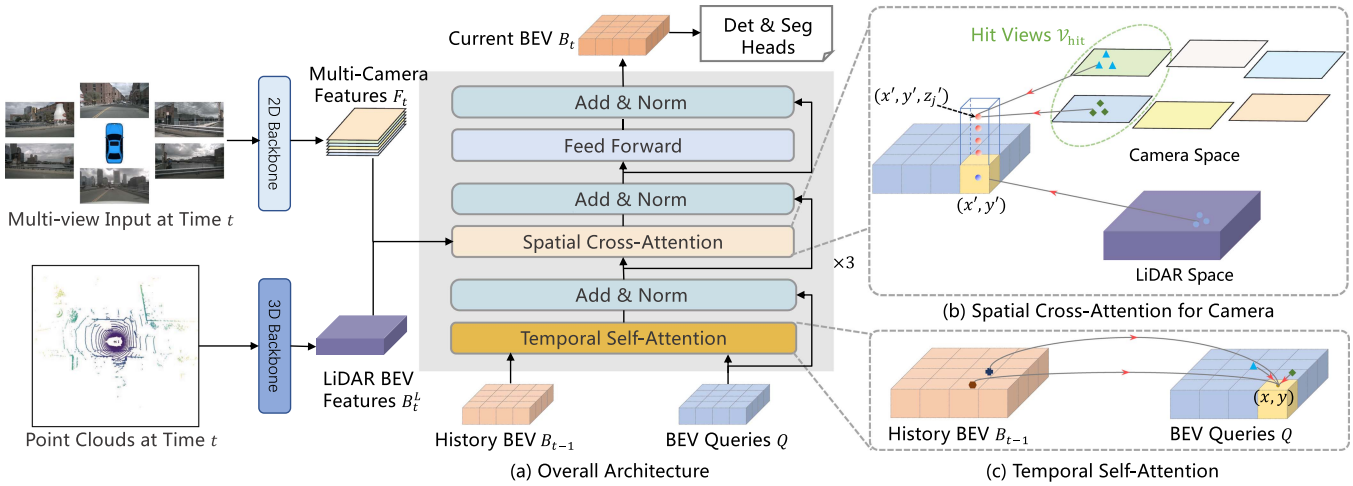


Fig. 3. Overall architecture of BEVFormer. (a) The encoder layer of BEVFormer contains grid-shaped BEV queries, temporal self-attention, and spatial cross-attention. (b) In spatial cross-attention, each BEV query only interacts with spatial features in the regions of interest. (c) In temporal self-attention, each BEV query interacts with two features: the BEV queries at the current timestamp and the BEV features at the previous timestamp.

and point cloud data as well as temporal features from history BEV, according to the BEV query.

During inference, at timestamp t, we feed multi-camera images to the image backbone network (e.g., ResNet-101 [55]), and obtain the camera features $F_t = \{F_t^i\}_{i=1}^{N_{\text{view}}}$ of different camera views, where F_t^i is the feature of the ith view, N_{view} is the total number of camera views. At the same time, we also feed point clouds to the LiDAR backbone to get a LiDAR BEV representation B_t^L directly. Note that we preserved the BEV features B_{t-1} at the prior timestamp $t-1$. In each encoder layer, we first use BEV queries Q to query the temporal information from the prior BEV features B_{t-1} via the temporal self-attention. We then employ BEV queries Q to inquire about the spatial information from the multi-camera features F_t, and LiDAR BEV features B_t^L via the multi-modality spatial cross-attention. After the feed-forward network [54], the encoder layer output the refined BEV features, which is the input of the next encoder layer. After several stacking encoder layers, unified BEV features B_t at the current timestamp t are generated. Taking the BEV features B_t as input, the 3D detection head and map segmentation head predict the perception results, such as 3D bounding boxes and semantic maps.

B. BEV Queries

We predefine a group of grid-shaped learnable parameters $Q \in \mathbb{R}^{H \times W \times C}$ as the queries of BEVFormer, where H, W are the spatial shape of the BEV plane. To be specific, the query $Q_p \in \mathbb{R}^{1 \times C}$ located at $p = (x, y)$ of Q is responsible for the corresponding grid cell region in the BEV plane. Each grid cell in the BEV plane corresponds to a real-world size of s meters. The center of BEV features corresponds to the position of the ego car by default. Following common practices [56], we add learnable positional embedding to BEV queries Q before inputting them to BEVFormer.

C. Multi-Modality Spatial Cross-Attention

Since our BEV queries should interact with two different modalities (i.e., RGB images and point clouds), our multi-modality spatial cross-attention (MSCA) consists of two parts. The first part is for mining spatial information from camera data, and the other is for extracting information from LiDAR input. The whole process of MSCA can be formulated as follows:

$$\text{MSCA}(Q_p, B_t^L, F_t) = \alpha \cdot \text{SCA}_L(Q_p, B_t^L) \tag{2}$$

$$+ (1 - \alpha) \cdot \text{SCA}_C(Q_p, F_t), \tag{3}$$

where $\text{MSCA}(\cdot)$ denotes the multi-modality spatial cross-attention, and $\text{SCA}_L(\cdot)$ and $\text{SCA}_C(\cdot)$ are the spatial cross-attention for LiDAR and camera respectively. α is a learnable parameter to balance the weights of different modalities and is initialized with 0.5.

Spatial Cross-attention for LiDAR: Since the BEV queries and LiDAR BEV features have no perspective differences, we can directly adopt a cross-attention and extract the required information from LiDAR BEV features into each BEV query. To reduce the computational cost, we use sparse deformable attention [20], which is a resource-efficient attention layer. With the sparse attention mechanism, the BEV query Q_P is only required to interact with several sampling points around the point p in the LiDAR BEV features. This process can be formulated as

$$\text{SCA}_L(Q_p, B_t^L) = \text{DAttn}\left(Q_p, B_t^L\right), \tag{4}$$

where Q_p is the BEV query and B_t^L is the LiDAR BEV features.

Spatial Cross-attention for Camera: Due to the large input scale of multi-camera 3D perception (containing N_{view} camera views), the computational cost of vanilla multi-head attention [54] is extremely high. Therefore, we continue to develop the spatial cross-attention for the camera based on deformable attention [20], where each BEV query Q_p only interacts with its regions of interest across camera views. However, deformable

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attention is originally designed for 2D perception, so some adjustments are required for 3D scenes.

As shown in Fig. 3(b), we first lift each query on the BEV plane to a pillar-like query [1], sample N_{ref} 3D reference points from the pillar, and then project these points to 2D views. For one BEV query, the projected 2D points can only fall on some views, and others are not hit. Here, we term the hit views as $\mathcal{V}_{\text{hit}}$. After that, we regard these 2D points as the reference points of the query Q_p and sample the features from the hit views $\mathcal{V}_{\text{hit}}$ around these reference points. Finally, we perform a weighted sum of the sampled features as the output of spatial cross-attention. The process of spatial cross-attention for the camera can be formulated as follows:

$$\text{SCA}_C(Q_p, F_t) = \frac{1}{|\mathcal{V}_{\text{hit}}|} \sum_{i \in \mathcal{V}_{\text{hit}}} \sum_{j=1}^{N_{\text{ref}}} \text{DAttn}\left(Q_p, \mathcal{P}(p,i,j), F_t^i\right), \tag{5}$$

where i indexes the camera view, j indexes the reference points, and N_{ref} is the total reference points for each BEV query. F_t^i is the features of the ith camera view. For each BEV query Q_p, we use a project function $\mathcal{P}(p,i,j)$ to get the jth reference point on the ith view image.

Next, we introduce how to obtain the reference points on the view image from the projection function $\mathcal{P}$. We first calculate the real-world location (x', y') corresponding to the query Q_p located at $p = (x, y)$ of Q as (6).

$$x' = \left(x - \frac{W}{2}\right) \times s; \quad y' = \left(y - \frac{H}{2}\right) \times s, \tag{6}$$

where H, W are the spatial shape of BEV queries, the size of the resolution of BEV's grids is s, and coordinates (x', y') of ego car is the origin. In 3D space, the objects located at (x', y') will appear at the height of z' on the z-axis. So we predefine a set of anchor heights $\{z'_j\}_{j=1}^{N_{\text{ref}}}$ to make sure we can capture clues that appeared at different heights. In this way, for each query Q_p, we obtain a pillar of 3D reference points $(x', y', z'_j)_{j=1}^{N_{\text{ref}}}$. Finally, we project the 3D reference points to different image views through the projection matrix of cameras, which can be written as

$$\mathcal{P}(p,i,j) = (x_{ij}, y_{ij})$$
$$\text{where } z_{ij} \cdot \begin{bmatrix} x_{ij} & y_{ij} & 1 \end{bmatrix}^T = T_i \cdot \begin{bmatrix} x' & y' & z'_j & 1 \end{bmatrix}^T. \tag{7}$$

Here, $\mathcal{P}(p,i,j)$ is the 2D point on ith view projected from jth 3D point (x', y', z'_j), $T_i \in \mathbb{R}^{3\times4}$ is the known projection matrix of the ith camera.

Global attention: Besides deformable attention [20], our spatial cross-attention for the camera can also be implemented by other attention mechanisms, such as vanilla multi-head attention [54]. The most straightforward way to employ global attention is making each BEV query interact with all multi-camera features, and this conceptual implementation does not require camera calibration. However, the computational cost of this straightforward way is unaffordable. Therefore, we still utilize the camera's intrinsic and extrinsic to decide the hit views that one BEV query deserves to interact with. This strategy makes that one BEV query usually interacts with only one or two views rather than all views, making it possible to use global attention in spatial cross-attention. Notably, compared to other attention mechanisms that rely on precise camera intrinsic and extrinsic, global attention is more robust to camera calibration, and detailed results will be reported in our experiment section. By default, considering the model performance and computational cost, using deformable attention is our optimal solution.

D. Temporal Self-Attention

In addition to spatial information, temporal information is also crucial for the perception system to understand the surrounding environment [14]. For example, it is challenging to infer the velocity of moving objects or detect highly occluded objects from static images without temporal clues. Although current LiDAR-based methods usually stack multiple point could sweeps to density the inputs, thus implicitly encodes short-term temporal clues, it still lacks temporal clues from camera data and long-term temporal dependency. To address this problem, we design temporal self-attention, which can represent the current environment by incorporating history BEV features. Using the fused BEV features to conduct the interaction of temporal information does not need to consider a specific modality, which simplifies the pipeline.

Given the BEV queries Q at current timestamp t and history BEV features B_{t-1} preserved at timestamp $t-1$, we first align B_{t-1} to Q according to ego-motion to make the features at the same grid correspond to the same real-world location. Here, we denote the aligned history BEV features B_{t-1} as B'_{t-1}. However, from times $t-1$ to t, movable objects travel in the real world with various offsets. It is challenging to construct the precise association of the same objects between the BEV features of different times. Therefore, we model this temporal connection between features through the temporal self-attention (TSA) layer, which can be written as follows:

$$\text{TSA}\left(Q_p, \{Q, B'_{t-1}\}\right) = \sum_{V \in \{Q, B'_{t-1}\}} \text{DAttn}\left(Q_p, p, V\right), \tag{8}$$

where Q_p denotes the BEV query located at $p = (x, y)$. In addition, different from the vanilla deformable attention, the offsets Δp in temporal self-attention are predicted by the concatenation of Q and B'_{t-1}. Especially, for the first sample of each sequence, the temporal self-attention will degenerate into a self-attention without temporal information, where we replace the BEV features $\{Q, B'_{t-1}\}$ with duplicate BEV queries $\{Q, Q\}$.

Compared to simply stacking BEV in [57], [58], [59], our temporal self-attention can more effectively model long temporal dependency. BEVFormer extracts temporal information from the previous BEV features rather than multiple stacking BEV features, thus requiring less computational cost and suffering less disturbing information. We also adapt BEVFormer into a static model called *BEVFormer-S* via adjusting the temporal self-attention into a vanilla self-attention without using history BEV features. The vision-only model and multi-model fusion model that do not use temporal information are denoted as *BEVFormer-S-C* and *BEVFormer-S-M*, respectively.

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E. Fusion Methods for Comparison

We briefly introduce the implementation details of the other methods presented in Fig. 2. For the vision-centric fusion method, when performing 3D sampling in the Spatial Cross-attention for Camera, we only consider 3D sampling points where point cloud voxels exist, and other sampling points not occupied by point cloud voxels will be seen as void features. For LiDAR-centric fusion, we use point cloud coordinates as 3D sampling points to obtain corresponding image features through Spatial Cross-attention for Camera, serving as the semantic information for the point cloud. For the independent fusion strategy, we mainly follow BEVFusion-MIT [4] in concatenating the BEV features generated from the conference version of BEVFormer with BEV features derived from point clouds.

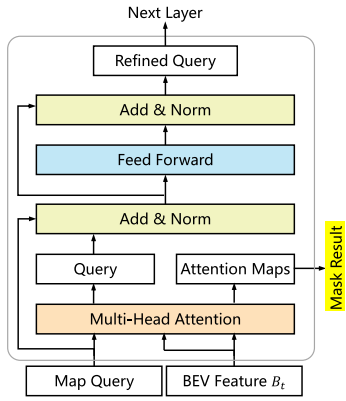


Fig. 4. Segmentation head (mask decoder) of BEVFormer.

F. Applications of BEV Features

Since the BEV features $B_t \in \mathbb{R}^{H\times W\times C}$ is a versatile 2D feature map that can be used for various autonomous driving perception tasks, the 3D object detection and map segmentation task heads can be developed based on 2D perception methods [20], [21] with minor modifications.

For 3D object detection, we design an end-to-end 3D detection head based on the 2D detector Deformable DETR [20]. The modifications include using single-scale BEV features B_t as the input of the decoder, predicting 3D bounding boxes and velocity rather than 2D bounding boxes, and only using L_1 loss to supervise 3D bounding box regression. With the detection head, our model can end-to-end predict 3D bounding boxes and velocity without the NMS post-processing. We predict 10 parameters for each 3D bounding box, including the 3 parameters (l, w, h) for the scale of each box, 3 parameters (x_o, y_o, z_o) for the center location, 2 parameters $(\cos(\theta), \sin(\theta))$ for object's yaw θ, 2 parameters (v_x, v_y) for the velocity. Only L_1 loss and L_1 cost are used during the training phase. Following [32], we use 900 object queries and keep 300 predicted boxes with the highest confidence scores during inference.

For 3D object tracking, inspired by MOTR [60], we have extended our detection head to object tracking task. Unlike MOTR, which only retains the track query from the previous frame, we maintain track query sequences over a longer temporal window, which helps in more robust tracking.

For map segmentation, we design a map segmentation head based on a 2D segmentation method Panoptic SegFormer [21]. Since the map segmentation based on the BEV is the same as the common semantic segmentation, we utilize the mask decoder of [21] and class-fixed queries to target each semantic category, including the car, vehicles, road (drivable area), and lane. As shown in Fig. 4, for each class of the semantic map, we follow the mask decoder in [21] to use one learnable query to represent this class and generate the final segmentation masks based on the attention maps from the vanilla multi-head attention.

For vectorized map construction, compared to map segmentation tasks, this task can provide richer road topology information. We built our own online map construction head based on MapTR [61]. Compared to MapTR, we propose a lane-aware self-attention mechanism that enables lane queries to focus more on adjacent lane information.

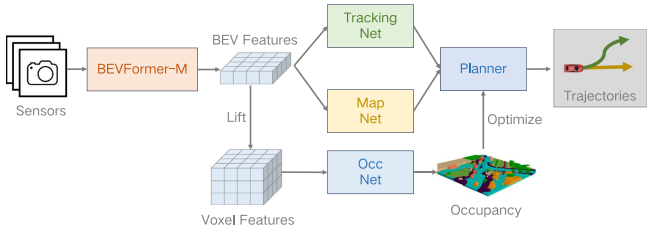


Fig. 5. Architecture of End-to-End Autonomous Driving. Tracking Net and Map Net are query-based networks. After the Planner generates initial prediction results, it utilizes the Occupancy results for optimization to avoid collisions.

For occupancy prediction, we use FB-OCC [62], [63] as our occupancy prediction head. Since our method generates a 2D BEV feature, we lift the BEV feature into 3D voxel features by adding embeddings to different heights.

For end-to-end autonomous driving, inspired by [23], [64], we feed the tracking queries and vectorized map queries to the planner and use the occupancy results to constrain the predicted trajectories, as shown in Fig. 5.

G. Implementation Details

Training Phase: For each sample at timestamp t, we randomly sample another 3 samples from the consecutive sequence of the past 2 seconds, and this random sampling strategy can augment the diversity of ego-motion [65]. We denote the timestamps of these four samples as $t-3$, $t-2$, $t-1$ and t. For the samples of the first three timestamps, they are responsible for recurrently generating the BEV features $\{B_{t-3}, B_{t-2}, B_{t-1}\}$ and this phase requires no gradients. For the first sample at timestamp $t-3$, there are no previous BEV features, and temporal self-attention degenerates into self-attention. At the time t, the model generates the BEV features B_t based on both multi-modality inputs and the prior BEV features B_{t-1}, so that B_t contains the temporal and spatial clues crossing the four samples. Finally, we feed the BEV features B_t into the detection and segmentation heads and compute the corresponding loss functions.

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Inference Phase: During the inference phase, we evaluate each frame of the video sequence in chronological order. The BEV features of the previous timestamp are saved and used for the next, and this online inference strategy is time-efficient and consistent with practical applications. Although we utilize temporal information, our inference speed is still comparable with other static methods [30], [32].

IV. Experiments

A. Datasets

We conduct experiments on the challenging public autonomous driving dataset, namely nuScenes dataset [3].

The nuScenes dataset [3] contains 1,000 scenes of roughly 20s duration each, and the key samples are annotated at 2Hz. Each sample consists of RGB images from 6 cameras and has 360° horizontal FOV and a 32-beam LiDAR scan. For the detection task, there are 1.4M annotated 3D bounding boxes from 10 categories. We follow the settings in [74] to perform the BEV map segmentation task. This dataset also provides the official evaluation metrics for the detection task. The mean average precision (mAP) of nuScenes is computed using the center distance on the ground plane rather than the 3D Intersection over Union (IoU) to match the predicted results and ground truth. The nuScenes metrics also contain 5 types of true positive metrics (TP metrics), including ATE, ASE, AOE, AVE, and AAE, for measuring translation, scale, orientation, velocity, and attribute errors, respectively. The nuScenes also defines a nuScenes detection score (NDS) as $\mathrm{NDS}=\frac{1}{10}[5\mathrm{mAP}+\sum_{\mathrm{mTP}\in\mathbb{TP}}(1-\min(1,\mathrm{mTP}))]$ to capture all aspects of the nuScenes detection tasks. For tracking task, we use the metrics of AMOTA (Average Multi-object Tracking Accuracy), AMOTP (Average Multi-object Tracking Precision), Recall. For map segmentation and occupancy tasks, we use mIoU (mean Intersection-over-Union) metric. For end-to-end autonomous driving, we use the commonly used L2 distance and collision rate metrics.

B. Implementation Details

1) Fusion Methods on Top of BEVFormer: In Vision-Centric Fusion, the point cloud only provides geometric information and does not require encoding through an encoder. Based on the features of BEVFormer, we use point cloud information to control the sampling points of BEVFormer in 3D space. In detail, we first voxelized the point cloud to obtain a simple binary occupancy mask in 3D space. We only see the 3D sampling point has a corresponding valid occupancy voxel that can further sample image features. Other sample points will be ignored. This can prevent 3D sampling points from collecting incorrect information, but at the same time, due to the sparseness of the point cloud, it may lead to the omission of important information.

For LiDAR-Centric Fusion methods, they only utilized image semantic masks to enhance the point cloud. In this paper, we employ high-level image features instead. Additionally, benefiting from SCA-C (Spatial Cross-Attention for Cameras), we can sample the image features of a local region rather than just a single point. In detail, for each point cloud, we project it to the images and use SCA-C to collect corresponding image features. Then these image features can just be seen as the properties of the corresponding point cloud.

For Independent Fusion, we largely follow the implementation of BEVFusion-MIT, except that we replace the camera stream with BEVFormer-C.

2) Downstream Tasks: Object Tracking For tracking methods such as MOTR and UniAD, they only keep the object queries from the previous moment at each point in time. We believe this is not the optimal choice. In our implementation, we retain a long-term window of historical object queries as memory. A longer temporal window allows for more stable tracking of objects, resulting in lower ID Switches (IDS).

Vectorized map construction: Our method, based on MapTR, made the following improvements: we removed point queries, allowing each lane query to directly predict the final s. When lane queries perform self-attention, we used a lane-aware self-attention mechanism to increase the weight between adjacent lanes. In detail, we can compute the MDE (min distance error) between each pair of lanes. The lane pair with lower MDE should have higher attention weight while performing self-attention.

Occupancy prediction: We adopt the occupancy task head of FB-OCC. To lift the BEV features from 2D to 3D voxel features, we repeat the BEV features along the high axis and add different learnable embeddings.

End-to-end autonomous driving: For our end-to-end autonomous driving model, the Tracking Net and the Map Net are the models introduced above. Since they are both the query-based method. The planner accepts the tracking query and the map query as input and generates the initial trajectories. To avoid collision with the objects that failed to be detected by the detection model, we use the predicted occupancy results to optimize the initial trajectories to avoid collision.

C. Experimental Settings

While conducting sensor-fusion experiments, for the LiDAR steam, we follow the commonly used settings [4], [5], [10] and use the voxel-based LiDAR backbone. For the camera stream, we use the ResNet-101-DCN [25], [55] that initialized from FCOS3D [30] checkpoint. By default, we utilize the output multi-scale features from FPN [75] with sizes of 116, 132, 164 and the dimension of $C=256$. The perception range is [− 54m, 54m] for the X and Y axis, the default size of BEV queries is 180×180, and the size of resolution s of BEV's grid is 0.6m. The BEV encoder contains 3 encoder layers and constantly refines the BEV queries in each layer. The input BEV features B_{t-1} for each encoder layer are the same and require no gradients. For each local query, during the spatial cross-attention module for the camera (SCA-C) implemented by deformable attention mechanism, it corresponds to $N_{\text{ref}}=4$ target points with different heights in 3D space, and the predefined height anchors are sampled uniformly from -5 meters to 3 meters. For each reference point on 2D view features, we use four sampling points around this reference point for each head. The pre-trained LiDAR stream is from TransFusion-L [10], which is commonly used in recent sensor fusion methods. We train our sensor-fusion model by

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TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE NUSCENES `Test` AND `Val` SET

Method	Modality	Backbones		*validation*		*test*	
		Image	LiDAR	mAP↑	NDS↑	mAP↑	NDS↑
DETR3D [32]	C	V99 [66]	-	-	-	41.2	47.9
PETR[33]	C	V99	-	-	-	44.1	50.4
BEVDet4D [37]	C	Swin-Base [67]	-	42.1	54.5	45.1	**56.9**
Ego3RT [68]	C	V99	-	47.8	53.4	42.5	47.3
BEVFormer-C	C	R101	-	41.6	51.7	44.5	53.5
BEVFormer-C	C	V99	-	44.6	54.2	**48.1**	**56.9**
CenterPoint [69]	L	-	VoxelNet	59.6	66.8	60.3	67.3
Focals Conv [70]	L	-	VoxelNet-FocalsConv	61.2	68.1	63.8	70.0
Transfusion-L [10]	L	-	VoxelNet	65.1	70.1	65.5	70.2
BEVFormer-L	L	-	VoxelNet	**66.3**	**70.3**	**66.7**	**70.4**
FUTR3D [8]	L+C	R101	VoxelNet	64.5	68.3	-	-
PointAugmenting [11]	L+C	DLA34 [71]	VoxelNet	-	-	66.8	71.0
MVP [51]	L+C	DLA34	VoxelNet	67.1	70.8	66.4	70.5
AutoAlignV2 [9]	L+C	CSPNet [72]	VoxelNet	67.1	71.2	68.4	72.4
TransFusion [10]	L+C	R50	VoxelNet	67.5	71.3	68.9	71.6
BEVFusion [5]	L+C	Swin-Tiny [67]	VoxelNet	67.9	71.0	69.2	71.8
BEVFusion [4]	L+C	Swin-Tiny	VoxelNet	68.5	71.4	70.2	72.9
BEVFormer-M	L+C	R101	VoxelNet	**71.2**	**73.2**	**72.9**	**74.1**

'L' and 'C' indicate LiDAR and camera, respectively.

loading the LiDAR steam and camera steam checkpoint for 10 epochs, a batch size of 2 per GPU, a learning rate of 1×10^{-4}, and we decay the learning rate with a cyclic annealing [76]. We employ AdamW [77] with a weight decay of 5×10^{-2} to optimize our models. For the data augmentation strategy, we follow the common practice and use random flipping, random rotating with a range of $[-\pi/4, \pi/4]$, and random scaling with a range of [0.95, 1.05]. We didn't apply test-time augmentation (TTA) and model ensemble since they seriously affected the efficiency of model inference.

In addition to the sensor fusion configuration, we also perform camera-only and LiDAR-only experiments. Compared to the sensor fusion model, the camera-only detector has 6 encoder layers, a BEV size of 200×200, and a powerful backbone VoVnet-99 [66]. In this setting, the spatial cross-attention module only extracts spatial information from the camera modality. For the LiDAR-only setting, it is the same as the sensor-fusion configuration, except the spatial cross-attention module only uses LiDAR BEV features as input.

D. 3D Object Detection Results

In this section, we report the detection performance of BEVFormer with different modalities and compare our results with previous state-of-the-art methods. In Table I, we report our main results on nuScenes `test` and `val` splits. Under the sensor fusion setting, BEVFormer-M obtains 74.1% NDS and 72.9% mAP, which outperforms the previous method BEVFusion [4] over 1.2% NDS on `test` set. For the camera-based method, BEVFormer-L performs slightly better than TransFusion-L [10]. Compared to the single-modality input BEVFormer-L, BEVFormer-M obtains a 3.9% mAP gain from modality complementary, which is 2.5% NDS higher than the gain obtained by TransFusion [10].

For a camera-only manner, BEVFofmer-C achieves 56.9% NDS, which outperforms the previous method DETR3D [32] over 9 points on `test`, under fair training strategy and comparable model scales. Previous camera-based methods [30], [32], [90] were almost unable to estimate the velocity, and our method demonstrates that temporal information plays a crucial role in velocity estimation for multi-camera detection. As shown in Table II, the mean Average Velocity Error (mAVE) of BEVFormer-C is 0.394m / s on the `val` set, outperforming other camera-based methods by a vast margin and approaching the performance of LiDAR-based methods.

E. Ablation of Fusion Strategies

As shown in Table III, we compare the object detection performance of methods based on different fusion strategies. We can observe that our proposed interactive fusion method achieved the best results among all fusion strategies. For the vision-centric fusion, we can observe that its results are significantly lower than others, mainly because the point cloud information only provides spatial guidance. Relying mainly on visual information, as opposed to other methods that use a LiDAR encoder to effectively encode point cloud information, is not a good fusion strategy. The LiDAR-centric fusion method performs better than BEVFormer-L, but still falls short of the Independent fusion strategy and the interactive fusion method. A reasonable explanation is that the sparse point clouds only utilize features around the projected points from the image information, and the images do not fully participate in the fusion process. Such a strategy is also suboptimal. Compared to Independent Fusion, our method facilitates a deeper level of fusion between modalities, rather than having each modality operate independently from one another. Our method outperforms the commonly used independent fusion strategy with 0.7% NDS. We wish to

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TABLE II
3D DETECTION RESULTS ON NUSCENES val SET

Method	Modality	NDS(%)↑	mAP(%)↑	mATE↓	mASE↓	mAOE↓	mAVE↓	mAAE↓
FCOS3D [30]	C	41.5	34.3	0.725	0.263	0.422	1.292	0.153
PGD [73]	C	42.8	36.9	0.683	0.260	0.439	1.268	0.185
DETR3D [32]	C	42.5	34.6	0.773	0.268	0.383	0.842	0.216
BEVFormer-S-C	C	44.8	37.5	0.725	0.272	0.391	0.802	0.200
BEVFormer-C	C	51.7	41.6	0.673	0.274	0.372	0.394	0.198
BEVFormer-S-M	L+C	72.0	69.8	0.275	0.252	0.257	0.318	0.187
BEVFormer-M	L+C	73.2	71.2	0.278	0.253	0.266	0.265	0.186

"C" indicates Camera and "L" indicates LiDAR.

TABLE III
3D DETECTION RESULTS OF DIFFERENT FUSION STRATEGY ON NUSCENES VAL SET

Method	NDS (%) ↑	mAP↑
BEVFormer-L	70.3	66.3
Vision-Centric Fusion	70.6	67.0
LiDAR-Centric Fusion	71.0	68.2
Independent Fusion	72.5	70.5
Interactive Fusion	73.2	71.2

TABLE IV
MULTI-OBJECT TRACKING

Method	split	AMOTA↑	AMOTP↓	Recall↑
ViP3D [78]	val	0.217	1.625	0.363
QD3DT [79]	val	0.242	1.518	0.399
MUTR3D [80]	val	0.294	1.498	0.427
UniAD [23]	val	0.359	1.320	0.467
BEVFormer-C	val	0.405	1.250	0.500
EagerMOT [81]	test	0.677	0.550	0.727
AlphaTrack[82]	test	0.693	0.585	0.723
UVTR [83]	test	0.701	0.686	0.750
BEVFormer-M	test	0.721	0.650	0.770

Compared to previous methods, the tracking model based on BEVFormer has a clear advantage.

TABLE V
OCCUPANCY PREDICTION

Method	Backbone	Image Size	mIoU↑
OccFormer [84]	R101	1600×900	21.9
TPVFormer [85]	R101	1600×900	27.8
SparseOcc [86]	R50	704×256	30.9
BEVFormer-C	R50	704×256	35.9
BEVFormer-M	R50 & VoxelNet	704×256	55.3

Compared to previous methods, BEVFormer has achieved a significant performance improvement.

TABLE VI
MAP SEGMENTATION RESULTS ON NUSCENES val SET

Method	BEV Segmentation (IoU)			
	Car	Vehicles	Road	Lane
Lift-Splat† [74]	32.1	32.1	72.9	20.0
FIERY† [57]	-	38.2	-	-
VPN*	31.0	31.8	76.9	19.4
Lift-Splat*	42.1	41.7	77.7	20.0
BEVFormer-C	44.8	44.8	80.1	25.7
BEVFormer-L	58.7	58.8	77.1	22.7
BEVFormer-M	64.2	64.2	86.1	34.9

*: We use VPN [87] and Lift-Splat [74] to replace our BEV encoder for comparison, and the task heads are the same. †: Results from their paper.

TABLE VII
VECTORIZED MAP CONSTRUCTION

Method	Modality	Backbone	Epochs	mAP↑
VectorMapNet [88]	C	R50	110	40.9
VectorMapNet [88]	C & L	R50 & PointPillar	110	45.2
MapTR-tiny [61]	C	R50	24	50.3
BEVFormer-C	C	R50	24	55.1
BEVFormer-M	C & L	R50 & VoxelNet	24	65.0

BEVFormer also achieves leading results in vectorized map construction.'L' and 'C' indicate LiDAR and camera, respectively.

TABLE VIII
END-TO-END AUTONOMOUS DRIVING

Method	Modality	Avg. L2(m) ↓	Avg. Colli.(%)↓
UniAD [23]	C	1.03	0.77
VAD [64]	C	1.25	1.09
BEVFormer-C	C	0.99	0.70
BEVFormer-M	C & L	0.95	0.68

The metric implementations are based on [89]. Compared to UniAD, our method achieves a lower collision rate, effectively enhancing the safety of the autonomous driving system. No ego status is used in these methods. 'L' and 'C' indicate LiDAR and camera, respectively.

emphasize that our method avoids over-engineering and the use of complex modules. We believe that for autonomous driving systems, methods with a simpler design may offer distinct advantages.

F. Downstream Task Results

3D Object Tracking: As shown in Table IV, the tracking model based on BEVFormer achieves leading results under both vision-only and multi-modality fusion. For example, BEVFormer-M outperforms UVTR-M [83] by 2% AMOTA.

Occupancy Prediction: We show our occupancy prediction results on Table V. BEVFormer-C outperforms SparseOcc with 5 points. Benefiting from the accurate 3D positioning provided by point clouds, the fusion model BEVFormer-M achieves enhanced results.

Map Segmentation: We also select two different BEV generation methods VPN [87] and Lift-Splat [74], for comparison. The image backbone and the task heads are the same as the BEVFomer for fair comparisons. For VPN, We employ the official codes in this work. Limited by the huge amount of parameters of MLP, it is difficult for VPN [87] to generate high-resolution BEV (e.g., 200× 200). To compare with VPN, in this work, we transform the single-scale view features into BEV with a low resolution of 50×50 via two view translation layers. For Lift-Splat, we enhance the camera encoder of Lift-Splat [74]

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TABLE IX
THE DETECTION RESULTS OF DIFFERENT METHODS WITH VARIOUS BEV ENCODERS ON NUSCENES `val` SET

Method	Attention	NDS(%)↑	mAP(%)↑	mATE↓	mAOE↓	#Param.	FLOPs	Memory
VPN* [87]	-	33.4	25.2	0.926	0.598	111.2M	924.5G	~20G
List-Splat* [74]	-	39.7	34.8	0.784	0.537	74.0M	1087.7G	~20G
BEVFormer-S-C†	Global	40.4	32.5	0.837	0.442	62.1M	1245.1G	~36G
BEVFormer-S-C‡	Points	42.3	35.1	0.753	0.442	68.1M	1264.3G	~20G
BEVFormer-S-C	Local	**44.8**	**37.5**	**0.725**	**0.391**	68.7M	1303.5G	~20G

In order to verify the effect of our spatial cross-attention for camera, we choose BEVFormer-S-C to exclude the impact of temporal information and LiDAR input. "Memory" is the consumed GPU memory during training. *: We use VPN [87] and Lift- Splat [74] to replace BEV encoder of our model for comparison. †: We train BEVFormer-S-C using global attention in spatial cross-attention, and the model is trained with fp16 weights. In addition, we only adopt single-scale features from the backbone and set the spatial shape of BEV queries to be 100×100 to save memory. ‡: We degrade the interaction targets of deformable attention from the local region to the reference points only by removing the predicted offsets and weights.

with two additional convolutional layers for a fair comparison with our BEVFormer-C under a comparable parameter number. Other settings remain unchanged.

While comparing different modality inputs under the same settings as shown in Table VI, BEVFormer-M achieves higher performances than others. For example, BEVFormer-M outperforms BEVFormer-C by IoU of 9.2 points on lane segmentation (34.9% *versus* 25.7%).

Vectorized Map Construction: We present our vectorized map construction results in Table VII. Our method surpasses MapTR [61] by 5 points.

End-to-End Autonomous Driving: As shown in Table VIII, our proposed simple end-to-end autonomous driving method outperforms UniAD [23] in terms of L2 distance and collision rate.

G. Ablation Study

To delve into the effect of different modules, we conduct ablation experiments on nuScenes `val` set on detection task.

Effectiveness of Spatial Cross-Attention: To verify the effect of spatial cross-attention, we use the static version of BEVFormer to perform ablation experiments to exclude the interference of temporal information.

First we compare the different designs of spatial cross-attention for cameras and the results are shown in Table IX. The default spatial cross-attention for the camera is based on deformable attention. For comparison, we also construct two other baselines with different attention mechanisms: (1) Using the global attention to replace deformable attention; (2) Making each query only interact with its reference points rather than the surrounding local regions, and it is similar to previous methods [39], [40]. For a broader comparison, we also replace the BEVFormer with the BEV generation methods proposed by VPN [87] and Lift-Spalt [74]. We can observe that deformable attention significantly outperforms other attention mechanisms under a comparable model scale. Global attention consumes too much GPU memory, and point interaction has a limited receptive field. Sparse attention achieves better performance because it interacts with a prior determined region of interest, balancing receptive field and GPU consumption.

Then we also verify the effectiveness of spatial cross-attention for LiDAR. We design two other fusion strategies to perform the interaction between BEV queries and LiDAR BEV features. The first one concatenates two features and uses a 3×3 convolution to reduce the channel dimension, and the second one is simply adding them since they own the same tensor shape. As shown in Table X, we can observe that using deformable attention in spatial cross-attention for LiDAR achieves the best results and outperforms other strategies over 0.4% NDS.

Effectiveness of Temporal Self-Attention: We show the ablation results of using temporal self-attention in BEVFormer in Table II. With only camera modality, we can observe that BEVFormer-C outperforms BEVFormer-S-C with remarkable improvements under the same setting. For multi-modality input, since LiDAR input involves temporal information through stacking multiple point cloud sweeps, the gain brought by temporal self-attention is not as large as that of a pure camera setting. We still obtain a significant improvement of 1.2 points, which indicates that temporal information is beneficial for sensor fusion. The effect of temporal information is mainly in the following aspects: (1) The introduction of temporal information greatly benefits the accuracy of the velocity estimation; (2) The predicted locations and orientations of the objects are more accurate with temporal information; (3) We obtain higher recall on heavily occluded objects since the temporal information contains past objects clues, as showed in Fig. 6. To evaluate the performance of BEVFormer on objects with different occlusion levels, we divide the validation set of nuScenes into four subsets according to the official visibility label provided by nuScenes. In each subset, we also compute the average recall of all categories with a center distance threshold of 2 meters during matching. The maximum number of predicted boxes is 300 for all methods to compare recall fairly. On the subset that only 0-40% of objects can be visible, the average recall of BEVFormer-C outperforms BEVFormer-S-C and DETR3D with a margin of more than 6.0%, which shows the importance of temporal information for vison-only models. However, compared to vison-only models, the fusion model BEVFormer-M demonstrates overall advantages, especially for detecting occluded objects.

Effect of the Frame Number During Training: Table XI shows the effect of the frame number during training. We see that the NDS on nuScenes val set keeps rising with the growth of the frame number and begins to level off the frame number ≥ 4 for both camera modality and camera-LiDAR fusion. Therefore, we set the frame number during training to 4 by default in experiments.

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TABLE X
THE DETECTION RESULTS OF USING DIFFERENT INTERACTION STRATEGIES BETWEEN BEV QUERIES AND LIDAR BEV FEATURES

Method	SCA_L	NDS(%)↑	mAP(%)↑	mATE↓	mASE↓	mAOE↓	mAVE↓	mATE↓
BEVFormer-S-M	Add	71.6	69.2	0.284	0.253	0.262	0.310	0.188
BEVFormer-S-M	Conv.	71.6	69.5	0.281	0.256	0.260	0.332	0.190
BEVFormer-S-M	DAttn.	**72.0**	**69.8**	**0.275**	**0.252**	**0.257**	**0.318**	**0.187**

"Add" is simply adding two features. "Conv." is concatenating two features and using another 3×3 conversation layer to reduce channel dimension. "DAttn." indicates using deformable attention, which is our default strategy.

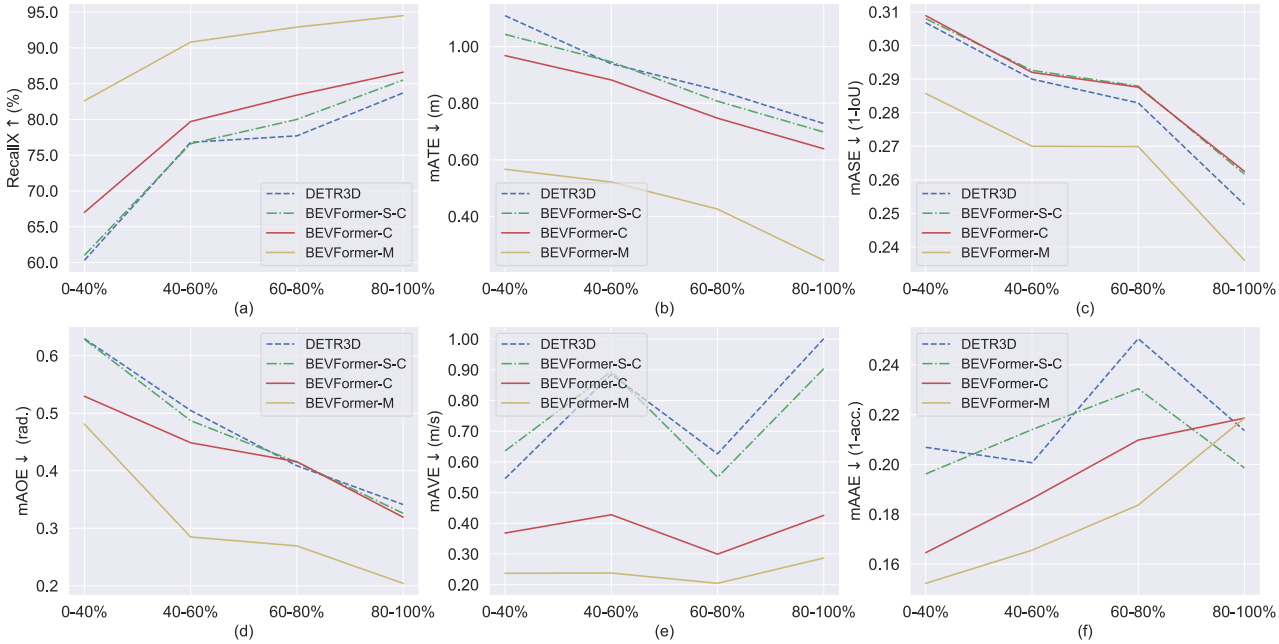


Fig. 6. The detection results of subsets with different visibilities. We divide the nuScenes `val` set into four subsets based on the visibility that {0-40%, 40-60%, 60-80%, 80-100%} of objects can be visible. (a): Enhanced by the temporal information, BEVFormer-C has a higher recall on all subsets, especially on the subset with the lowest visibility (0-40%) compared to BEVFormer-S-C. The fusion model BEVFormer-M gets much higher recall for occluded objects compared to the vision-only model BEVFormer-C. (b), (d) and (e): For vision-only models, temporal information benefits translation, orientation, and velocity accuracy. We also can observe that the fusion model BEVFormer-M has no obvious advantage in predicting the orientation of occluded objects. (c) and (f): For vison-only models, the scale and attribute error gaps among different methods are minimal. Temporal information does not work to benefit an object's scale prediction.

TABLE XI
NDS OF MODELS ON NUSCENES VAL SET USING DIFFERENT FRAME NUMBERS DURING TRAINING

#Frame	BEVFormer-C		BEVFormer-M	
	NDS(%)↑	mAP(%)	NDS(%)↑	mAP(%)↑
1	44.8	37.5	72.0	69.8
2	49.0	38.8	72.5	70.8
3	51.0	41.0	72.8	71.0
4	**51.7**	**41.6**	**73.2**	**71.2**
5	**51.7**	41.2	73.1	**71.2**

"#Frame" denotes the frame number during training.

TABLE XII
ABLATION EXPERIMENTS ON NUSCENES VAL SET

#	A.	R.	B.	C.	BEVFormer-C		BEVFormer-M	
					NDS(%)↑	mAP(%)	NDS(%)↑	mAP(%)↑
1	✗	✓	✓	✗	51.0	41.0	72.2	70.4
2	✓	✗	✓	✗	51.3	41.0	72.8	71.0
3	✓	✓	✗	✗	51.3	40.4	72.6	70.7
4	✓	✓	✗	✓	51.0	40.8	72.3	70.6
5	✓	✓	✓	✗	**51.7**	**41.6**	**73.2**	**71.2**

"A." indicates aligning history BEV features with egomotion. "R." indicates randomly sampling 4 frames from 5 continuous frames. "B." indicates using both BEV queries and history BEV features to predict offsets and weights. "C." indicates using convolutional layer rather than deformable attention.

Effect of Key Designs in Temporal Self-Attention: In temporal self-attention, we have several key designs for the final performance, including aligning history BEV features with ego-motion, randomly sampling history frames, and using both history features and current features to predict offsets. In addition to these designs, we also simply stack history features and current features and use one 3×3 convolutional layer to perform feature fusion to verify the effect of our temporal self-attention. Table XII shows the results of these ablation studies. Comparing #1 and #5, we see that aligning history BEV features with ego-motion is important to represent the same geometry scene as current BEV queries, which brings 1.0% NDS for the multi-modality model. Comparing #2 and #5, randomly sampling 4 frames from 5 frames is an effective data augment strategy to improve performance, which obtains a gain of 0.4% NDS for the multi-modality model. Compared to only

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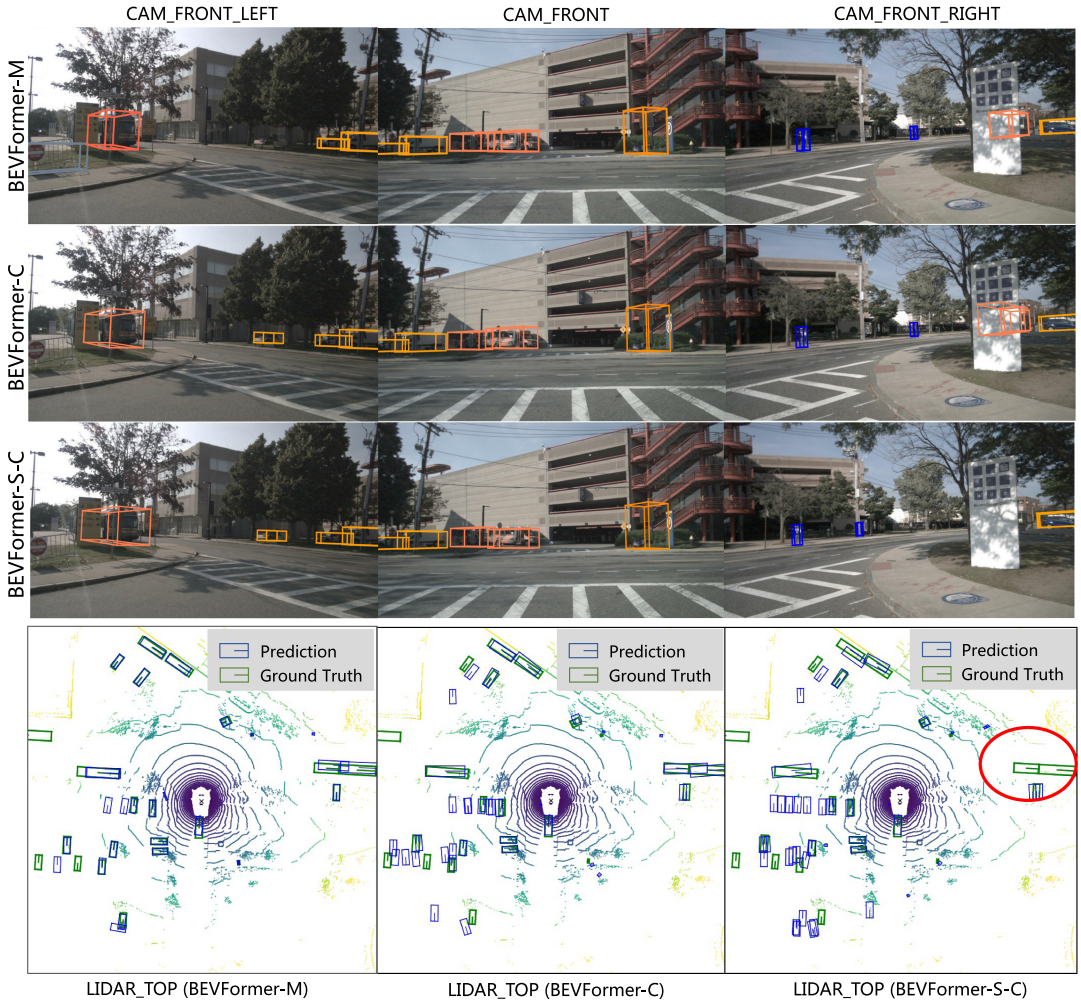


Fig. 7. Comparision of BEVFormer-M, BEVFormer-C and BEVFormer-S-C on nuScenes val set. We can observe that both BEVFormer-M and BEVFormer-C can detect highly occluded objects, and these objects are missed in the prediction results of BEVFormer-S-C (in red circle).

using the BEV query to predict offsets and weights during the temporal self-attention module (see #3), using both BEV queries and history BEV features (see #5) contain more clues about the past BEV features and benefits location prediction (73.2% NDS *versus* 52.6% NDS for multi-modality input). Comparing #4 and #5, we can observe that using deformable attention rather than convolutional layer brings an improvement of 0.9% NDS for the multi-modality model.

H. Visualization Results

As shown in Fig. 7, we compare BEVFormer-M, BEVFormer-C with BEVFormer-S-C. With temporal information, BEVFormer-C successfully detected two buses occluded by boards, but the static version BEVFormer-S-C fails to detect them. The fusion model BEVFormer-M can not only detect occluded objects but produce more accurate detection results compared to the vision-only models. We can observe that BEVFormer-M produces results with fewer hallucinations. We also show both object detection and map segmentation results in Fig. 8, where we see that the detection results and segmentation results are highly consistent.

I. Robustness on Camera Extrinsics

BEVFormer relies on camera intrinsics and extrinsics to obtain the reference points on 2D views in its spatial cross-attention for camera. During the deployment phase of autonomous driving systems, extrinsics may be biased due to various reasons such as calibration errors, camera offsets, etc. As shown in Fig. 9, we show the results of models under different camera extrinsics noise levels. Compared to BEVFormer-S-C (point), BEVFormer-S-C utilizes the spatial cross-attention based on deformable attention [20] and samples features around the reference points rather than only interacting with the reference points. With deformable attention, the robustness of BEVFormer-S-C is stronger than BEVFormer-S-C(point). For example, with the noise level being 4, the NDS of BEVFormer-S-C drops 15.2% (calculated by $1 - \frac{0.380}{0.448}$), while the NDS of BEVFormer-S-C (point) drops 17.3%. Compared to

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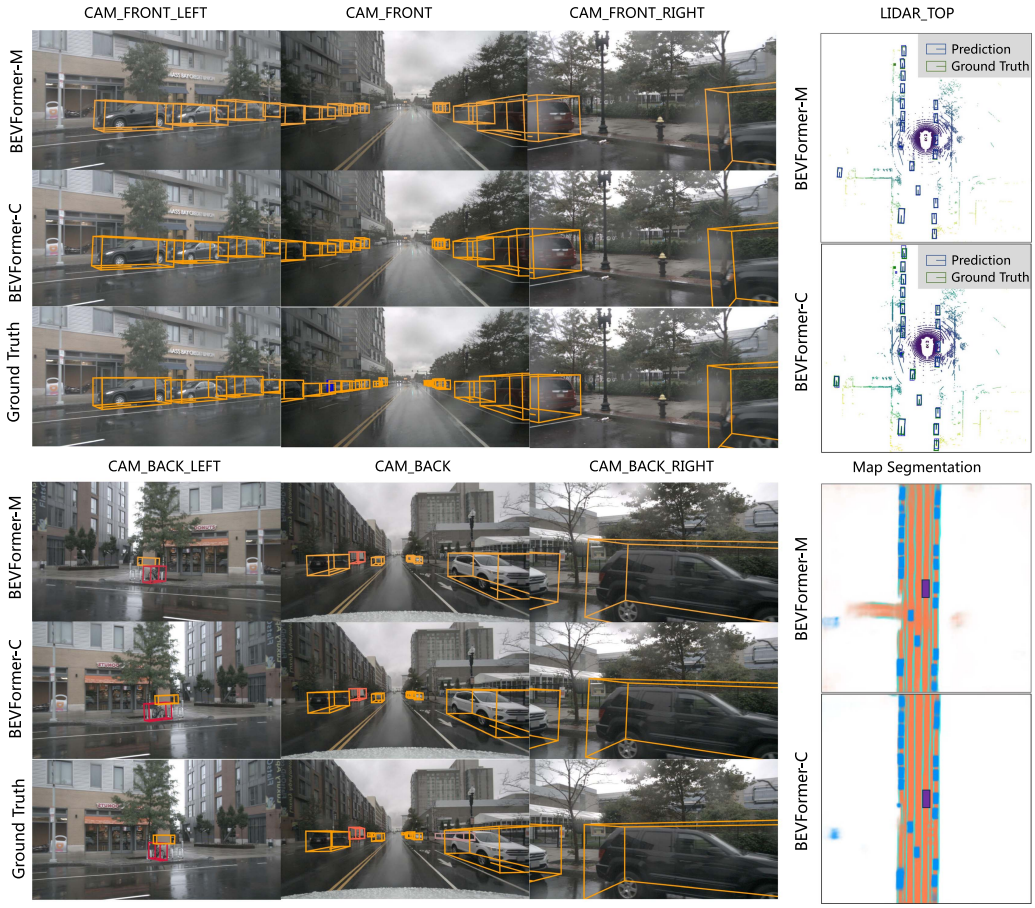


Fig. 8. Visualization results of both object detection and map segmentation tasks. We show vehicle, road, and lane segmentation in blue, orange, and green, respectively.

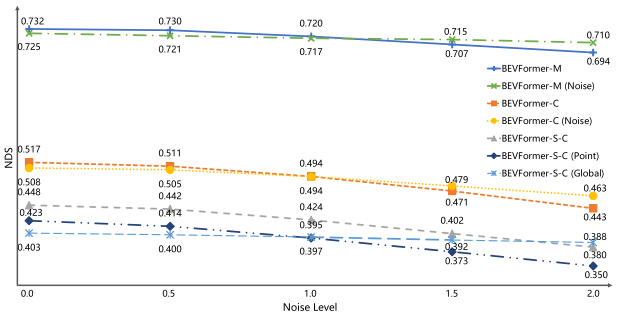


Fig. 9. NDS of methods on nuScenes val set subjected to different levels of camera extrinsics noises. For ith level noises, the rotation noises are sampled from a normal distribution with mean equals 0 and variance equals i (rotation noise are in degrees, and the noise of each axis is independent), and the translation noises are sampled from a normal distribution with mean equals 0 and variance equals $5i$ (translation noises are in centimeters, and the noise of each direction is independent). "BEVFormer-M" is our default fusion model and "BEVFormer-M (Noise)" is trained with noisy extrinsics under noise level 1. "BEVFormer-C" is our default version for camera-only input. "BEVFormer-C (noise)" is trained with noisy extrinsics (noise level=1). "BEVFormer-S-C" is our static version of BEVFormer-C with the spatial cross-attention implemented by deformable attention [20]. "BEVFormer-S-C (global)" is BEVFormer-S with the spatial cross-attention implemented by global attention (i.e., vanilla multi-head attention) [54]. "BEVFormer-S-C (point)" is BEVFormer-S-C with point spatial cross-attention where we degrade the interaction targets of deformable attention from the local region to the reference points only by removing the predicted offsets and weights.

BEVFormer-S-C, BEVFormer-C only drops 14.3% NDS, which shows that temporal information can also improve robustness on camera extrinsics. Following [74], we show that when training BEVFormer with noisy extrinsics, BEVFormer-C (noise) has stronger robustness (only drops 8.9% NDS). With the spatial cross-attention based on global attention, BEVFormer-S-C (global) has a strong anti-interference ability (4.0% NDS drop) even under level 4 of the camera extrinsics noise. The reason is that we do not utilize camera extrinsics to select the RoIs for BEV queries. Notably, under the harshest noises, we see that BEVFormer-S-C (global) even outperforms BEVFormer-S (38.8% NDS *versus* 38.0% NDS). For the fusion model BEVFormer-M, it also shows stronger anti-interference ability (3.8% NDS drop) benefiting from complementary multi-model information. Training under noise camera extrinsics, BEVFormer-M (Noise) only drops 1.5% points under the worst disturbance.

V. Discussion and Conclusion

In this work, we have proposed BEVFormer to generate the bird's-eye-view features from multi-modality inputs. BEVFormer can efficiently aggregate spatial and temporal information and generate powerful BEV features that simultaneously support 3D detection and map segmentation tasks. BEVFormer

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demonstrates that using spatiotemporal information from the multi-modality input can significantly improve the performance of 3D perception models. The advantages demonstrated by BEVFormer, such as more accurate velocity estimation and higher recall on low-visible objects, are essential for constructing a better and safer autonomous driving system and beyond.

V. Author Contributions

J. Dai leads the project, provides mentorship and allocates resources across task tracks. Z. Li, W. Wang, H. Li drafted the main outline of the project, worked on a preliminary version of the manuscript, and supervised key milestones of the project. Y. Qiao and T. Lu are the primary consultants, eliciting key goals and tracking progress. E. Xie, C. Sima oversaw the project from their inception.

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